PURDUE UNIVERSITY | Aeronautics and Astronautics
COLLEGE OF ENGINEERING

Course Information

Course number and title: **AAE 55000 – Multidisciplinary Design Optimization, Fall 2025**

CRN: **25218 (002) on-campus online students**
24280 (EPE) distance students in a program managed by Purdue Online-Engineering

Instructional Modality: **Asynchronous Online Learning**

Course credit hours: **3.000**

Public course webpage: https://engineering.purdue.edu/~crossley/aae550

Brightspace page: https://purdue.brightspace.com; look for Fall 2025 AAE 550 - Merge (All sections are merged into this Brightspace section.)

Prerequisites: Senior standing or graduate student. Knowledge of linear algebra and multivariate calculus. Computer programming skills sufficient to use functions available in Matlab. Some knowledge of basic statics and strength of materials might help with understanding examples and homework problems, but this knowledge is not required to succeed in the course. Appropriate equations and formulas for these problems will be provided.

Note: While Purdue's course numbering system changed a few years ago to use five digits, the TAs and I will refer to this course most often as AAE 550.

Instructor Information

Instructor: **Prof. William Crossley**

Course Description

AAE 550 is a fast-paced, graduate-level course that introduces students to the techniques of engineering design optimization, leading into topics needed for Multidisciplinary Design Optimization (MDO). The course also presents the application of these techniques to solve engineering design problems.

To accomplish these two tasks, AAE 550 has two overlapping parts. The first part of the course exposes students to basic concepts about and implementation of numerical optimization techniques, assuming that the student has little or no knowledge of these topics. The second part of the course uses this knowledge as the basis for students to investigate approaches for multiobjective and multidisciplinary optimization.

Learning Resources, Technology & Texts

Handouts / Notes

For each course module, I will provide a handout in Adobe Acrobat format that you will need to download from Brightspace, print (or download to your device, assuming you can "write" onto

the handout) and then have with you as you watch the corresponding lecture capture videos. These handouts are intentionally incomplete; during the videos, we will add information to these slides. My idea is to have you participate by taking notes during the video, because the activity of taking notes is associated with better retention of the material. However, with some of the material already provided, you should not need to rush to keep up with the notes and can listen more intently to the video.

These handouts are copyrighted by Prof. William A. Crossley, Purdue University and are for the use of students registered in AAE 550. Notes taken in class are generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern (**Course Hero for example**), for a course without the express written permission of the course instructor.

Textbooks

Consistent student feedback over many offerings of AAE 550 indicated that students felt the lecture discussions and handout material was sufficient to succeed in the course and that a textbook was not required. With the fast pace of the course covering many topics in a short time and with the graduate student target audience, a good reference for "self-teaching" may help. The instructor suggests that students acquire the following recommended text and use it to reinforce the information presented during the lectures.

Introduction to Optimum Design, Arora, J. S., Fifth Edition, Elsevier Academic Press, San Diego, CA, 2024. (ISBN: 9780-12-818320-5)

> This book discusses many, but not all, of the topics presented during the AAE 550 lectures. The book also presents the use of MATLAB and Excel for solving optimization problems. The fifth edition is the currently available print edition, and it has several improvements, but either the fourth or fifth edition will be fine as a reference for the course. Both the fifth and fourth editions are also available as a PDF format e-book. Information about the book is available from the publisher: https://shop.elsevier.com/books/introduction-to-optimum-design/arora/978-0-12-818320-5.
>
> The Purdue Libraries provide access to this text via Science Direct. You can start at https://www.lib.purdue.edu/ and enter "Introduction to Optimum Design" in the *Find Articles, Books, Media, Journals, Collections & Archives* field on the main page.

Required Software and Web Resources

MATLAB (Student Edition is acceptable) with Optimization Toolbox

> This is the most used software and optimization package in AAE 550, and you are required to have this software available. Purdue's current campus-wide MATLAB license allows Purdue students to download a copy of MATLAB to their own computer at no additional cost to the student. Visit: https://www.mathworks.com/academia/tah-portal/purdue-university-31484706.html and click on the "Sign in to get started" link. You need to have (or create) a mathworks.com account using your purdue.edu email address.

The MATLAB student version comes with the Optimization Toolbox, along with several other popular toolboxes. If you have access to the professional version of MATLAB with the Optimization Toolbox, that is also acceptable. Please download and use version R2025a.

Other Purdue options to use MATLAB are available via goremote or AppsAnywhere, but these have not been reliable enough to ensure assignment completion on time (i.e., use at your own risk); hence, the requirement to have your own copy of MATLAB with the Optimization Toolbox.

Microsoft Excel with Solver add-in

Microsoft Excel comes with the Solver add-in, which provides optimization routines for use within Excel spreadsheets. You must activate this add-in, if you have not used it before. Instructions to activate Solver are in the Excel help, or on the Microsoft Office Assistance web page. For instructions to activate the Solver add-in for Excel, visit https://support.office.com/en-us/article/Load-the-Solver-Add-in-in-Excel-612926fc-d53b-46b4-872c-e24772f078ca.

Microsoft Office, which includes Excel, is free for all students, visit https://www.itap.purdue.edu/shopping/software/product/office365.html

To date, we have not found a compatible solver product for Open Office or Google Sheets that is reliable for the assignments that use Solver in Excel. The underlying algorithms in the solver-like products for Open Office are not the same as those included in the Excel Solver, and the Frontline Systems Solver for Google Sheets does not appear to be reliable enough to use because the solver resides on Frontline Systems backend server, which has capacity limitations.

Brightspace learning management system

We will make use of Brightspace (https://purdue.brightspace.com) to provide handouts to accompany lectures, to post the recorded lectures, to post homework assignments and associated files, and to assign and automatically grade assessments and select parts of the homework assignments.

Gradescope

We will use Gradescope (https://www.gradescope.com/courses/1087763) to collect and grade your homework assignments. You will also be able to access the AAE 550 Gradescope page via a link in Brightspace.

Piazza

We will use Piazza (https://piazza.com/purdue/fall2025/wl202610aae55000epe/info) for discussion about AAE 550. There is a link to the AAE 550 Piazza class available in Brightspace. **Piazza should be your first option when asking questions about assignments** rather than sending email to the aae550@ecn.purdue.edu address. I encourage students to post responses when they can help other students. This will allow you to get help from classmates, the teaching assistants, and me. Piazza also has a good

LaTeX-based equation editor, which is important for AAE 550. We do request that you do not directly post answers to the assignment or homework on the discussion board. For instance, asking for clarification of method or a step is good, but we do not want you posting: "The answer I got for question 3 is 2.45 cm."

Learning Outcomes

In this course, you will:

1. Acquire basic knowledge about optimization techniques
2. Demonstrate familiarity with techniques for engineering design optimization
3. Demonstrate understanding in which methods are appropriate for a given optimization application
4. Develop the ability to formulate engineering problems as optimization problems that are appropriate for a chosen method
5. Utilize available computational tools and software to solve engineering design optimization problems

Teaching Philosophy

My approach to teaching engineering design optimization relies upon having students: (1) practice formulating problems, (2) examine important features of various optimization algorithms, and (3) use computer tools to solve optimization problems. This does not lend itself well to traditional exams, so the grade for AAE 550 will rely upon assessments, and homework assignments.

Assignments

The two types of assignments in AAE 550 are **assessments** and **homework**. Due dates for these will be available in the course schedule below and on Brightspace. Typically, assignments will be due on Tuesdays, except for weeks with holidays on Monday or Tuesday. There will be an assignment due every week starting on 03 September 2025.

The **assessments** are a group of occasionally two, but usually three or four, questions that use Brightspace's quiz tool; these will be graded automatically, and they take the place of questions that you might find at the end of a textbook chapter to test your knowledge of how various optimization algorithms work. There will be eight graded assessments planned for the semester. There will be a ninth assessment (titled Assessment 0) early in the semester that is not included in your grade; this lets you get practice using the Brightspace quiz tool before these affect your grade. Make sure that you click submit in the Brightspace quiz for each assessment before the due date, as the saved answers will not be automatically submitted on the due date. Each question in an assessment group will have equal weight in contributing to the grade for the assessment.

The **homework** assignments will ask you to pose and solve optimization problems using MATLAB and / or Excel as appropriate. You will be asked to make observations about your experience solving these optimization problems and compare this to concepts presented in class to

demonstrate that you have reached a high-level in meeting the educational objectives and learning goals for these topics. These are long and have multiple parts; they cannot be completed the night before they are due. There will be five graded homework assignments and an initial, un-graded homework 0. The five graded assignments are analogs to take-home exams for AAE 550 in terms of their contribution to your grade for the course.

Grading Scale

The final course grades will use the following weighting:

Items	Weighting
Assessments 1-8	36%
Homework 1-4	48%
Homework 5	16%

The final grades will use the plus/minus letter scheme. The following table converts the numerical percentage score, rounded to the nearest integer value (e.g., 91.55% becomes 92%, 91.45% becomes 91%) to the corresponding letter grades.

Score	Grade	Score	Grade	Score	Grade	Score	Grade
98 to 100%	A+	88 to 89%	B+	78 to 79%	C+	68 to 69%	D+
93 to 97%	A	83 to 87%	B	73 to 77%	C	63 to 67%	D
90 to 92%	A-	80 to 82%	B-	70 to 72%	C-	60 to 62%	D-

Each assessment and homework will be graded on a (points scored) / (points available) basis. The total points available will vary, but each assessment and homework will have equal weights as other assessments and homework assignments, except for Homework 5 (worth 16% rather than 12%). Graded assignments will use the criterion (straight-scale) approach shown above, but the instructor reserves the right to curve the grades if appropriate. Under no circumstance will the scale be more stringent than the criterion given below (e.g., 93% or above will always earn an A), and the curve will never span more than one grade scale (e.g., the lowest A possible when grades are curved is 83%). A total score of 50% or lower will always fail.

Course Schedule

The course is arranged around “modules” that encompass related topical areas. The schedule below provides suggested completion dates for viewing the content in each module, along with assignment deadlines.

Module	Topics
01	Class Introduction
02	Basic Concepts: Definitions, Optimal Design Problem Statement, Iterative Procedure, Solution Existence and Uniqueness, Kuhn-Tucker Conditions
03	Functions of One Variable – Concepts, Newton's Method, and Quadratic Fit, Golden Section Search, Matlab’s `fminbnd`

04	Unconstrained Functions in *N* Variables: First-Order Methods – Calculating Derivatives, Steepest Descent, Conjugate Direction, Variable Metric (DFP, BFGS); Second-Order Methods - Newton's Method, Variable Transformation and Scaling; Convergence Criteria, Matlab's `fminunc`, Excel Solver Add-in
05	Linear & Quadratic Programming: Simplex Method Excel Solver for LP, Matlab's `linprog`, Quadratic Programming, Matlab's `quadprog`
06	Constrained Functions in *N* Variables – Direct Methods: Introductory Concepts, Sequential Linear Programming (SLP), Method of Centers (MoC), Generalized Reduced Gradient (GRG) , Excel Solver Add-in, Sequential Quadratic Programming (SQP), Matlab's `fmincon`
07	Constrained Functions in *N* Variables – SUMT: Exterior and Interior Penalty Methods, Comparison of Penalty Methods, Constraint Scaling, Augmented Lagrange Multiplier (ALM) Method
08	Non-smooth and Global Optimization: Nelder-Mead Simplex, Simulated Annealing, Genetic Algorithms
09	Multiobjective Design: Pareto Optimality and Dominance, Weighted Sum, Weighted Sum, ε-Constraint (Gaming) Approach, Min-Max Approach, Goal Attainment
10	Approximation Techniques I: Basic Concepts, Derivative Based, Response Surface Methods, Least squares for Response Surface Coefficients, Design of Experiments, Global Response Surface Approximation, Sequential Response Surface Approximations
11	Multidisciplinary Optimization I: Decomposition Strategies, Multidisciplinary Feasible and All-at-Once Examples, Collaborative Optimization Example
12	Revisiting the AAE 55000 "Toolbox" and Closing Comments